

Santhosh Kumar Kuppan

santhosh.gnat@gmail.com

+91 78801 07379

[Portfolio](#)

[LinkedIn](#)

Aspiring mechanical engineer with a passion for fixed-wing aircraft, I design, simulate, and build things that fly.

Education:

B.Tech in Mechanical Engineering (2023 – 2027) | CGPA – 8.24

Vellore Institute of Technology, Chennai.

Experience:

1) Engineering Intern, Godrej Consumer Products Limited (May 2025 – June 2025):

- Surveyed an operating RO/DM plant, collected field measurements and developed a 1:65 scale P&ID layout that was later used for plant documentation and displayed outside the plant.
- Developed a 3D CAD model of a soap manufacturing plant for IoT dashboard visualization and process monitoring.
- Studied end-to-end FMCG production workflows across 3 product lines (soaps, mosquito repellent, hair colour), gaining exposure to batch production and plant-floor operations.
- Through frequent plant visits, analysed the working of boilers, RO/DM water plants, ETP, air compressors, chillers, cooling tower and understood the utilities involved in production.

2) Team Captain, Team Aviators International (May 2024 – Present):

- Led a 50-member aeromodelling team, developing fixed-wing UAVs (Unmanned Aerial Vehicles).
- Led the team in SAE DDC 2026, a national-level marquee aeromodelling competition, and secured **AIR 3 out of 45 teams** in our category and received a cash prize of **₹30,000**.
- Managed **₹2.5 lakh** team finances including budget, procurement, manufacturing and competition logistics.
- Initiated 5S in workplace through sort, set in order, shine, standardize and sustain. Resulted in an organized workplace by the introduction of a standardized storage system.
- Established team documentation that contains technical knowledge, project management practices and other critical information for future team members, benefitting over 30 juniors.
- Performed FEA using SolidWorks static simulation to evaluate structural integrity of wing supporting iterative design improvements.
- Coordinated with vendors for components sourcing and manufacturing.
- Allocated design, analysis, simulation, testing, validation, manufacturing and administrative tasks to team members and ensured timely completion.
- Executed end-to-end product development cycles from CAD modelling through manufacturing, assembly, testing, validation and iterative redesign.

Projects:

1) A Fixed Wing UAV for SAE 2026 – Team Aviators International:

- Led aircraft redesign due to packaging constraints and aiming to increase payload, filed ECR (Engineering Change Request) and the redesign resulted in reduction of packaging box volume by **~13% (3 ft³ to 2.6 ft³)**, increased payload fraction by **~33% (0.60 to 0.80)** and reduced empty weight by **~14% (1.2 kg to 1 kg)**.
- Performed external-flow **CFD** simulations in **ANSYS Fluent** at 0° and 5° angle of attack to obtain lift and drag forces and compared simulated lift with calculated lift.
- Used Python to process CFD results, calculate lift and drag coefficients, and plot aerodynamic results.
- Designed a fixed wing UAV with 1 m wingspan using SolidWorks capable of performing flight mission with a high payload fraction.
- Prepared DFM-oriented CAD models and fabricated .dxf and .stl files for laser cutting balsa wood and FDM 3D printing using PLA, PETG and TPU for various onboard flight components such as payload compartment, motor mount and hinges of control surfaces.
- Secured **1st place in Design Excellence at SAE DDC 2026** for an optimized design that adheres to the problem statement.

Skills:

- Software & Simulation: SolidWorks (**CSWA**), SolidWorks Simulation (FEA), ANSYS Fluent (CFD).
- Design & Manufacturing: CAD Modelling, Assemblies, DFM, DfAM, Design Validation, Iterative Design.
- Fabrication: FDM 3D Printing (PLA, PETG, TPU), Laser Cutting, Prototype Manufacturing.
- Domain Knowledge: **Aerodynamics**, UAV Flight Testing, Project Management.
- Languages: English (Professional), Tamil (Native), Hindi (Conversational).